Enhancing Factor Models with Machine Learning-Based Fundamental Prediction: An Approach to Mitigate Information Lag

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Abstract

Your abstract.

1 Introduction

The pursuit of models that can accurately explain and predict cross-sectional stock returns is a fundamental endeavor in financial economics. Firm-specific fundamentals are widely recognized as key drivers of equity performance, yet the precise functional form and timing of this relationship remain subjects of intense investigation. Foundational models, such as the Fama-French five-factor model, are designed to use fundamental data (e.g., book value from the previous financial report), which share the same time period as excess return, to construct factors (Fama & French, 2015). This design creates a critical misalignment when designing the prediction factor model: the factors used to predict future returns in period t+1 are based on information available at time t, which may no longer accurately represent a firm's current or imminent financial state. This lag may reduce the model's interpretability and ability to capture the true, timely relationship between a company's evolving fundamentals and its subsequent stock performance. This research aims to bridge this gap by introducing a novel machine learning framework, thereby contributing to the advancement of more sophisticated and powerful empirical asset pricing models.

This study is structured around two sequential primary objectives. First, to empirically validate the superior explanatory power of contemporaneous fundamentals over lagged fundamentals using a linear regression framework. We hypothesize that models using current fundamentals will significantly outperform those using past fundamentals in terms of goodness-of-fit (R^2) and the statistical significance of fundamental factor coefficients. Second, and more critically, to develop and validate a two-stage non-linear prediction model that enhances return forecasting, which involves employing a Multilayer Perceptron (MLP) to generate accurate, non-linear forecasts of the entire fundamental vector and utilizing these forecasted fundamentals with control variables in a second MLP to predict excess returns. We aim to demonstrate that this proposed model significantly outperforms a direct forecasting benchmark in both in-sample and out-of-sample tests.

In this paper, Chapter 2 reviews the relevant literature on linear factor models and the application of machine learning in asset pricing. Chapter 3 details the research methodology, which includes data source, processing, base hypothesis testing and model structure. Then this paper will display the empirical results. Finally, Chapter 6 concludes by summarizing the key empirical findings, discussing their implications for academics and practitioners.

2 Literature Review

The evolution of factor models, from the CAPM to the Fama-French five-factor model, has established a clear link between firm fundamentals and stock returns. However, these traditional linear approaches are inherently limited by their inability to capture complex, non-linear dynamics and their reliance on lagged accounting information, which fails to reflect a firm's current state due to the gradual diffusion of information into prices. The application of machine learning (ML) offers a paradigm shift. Seminal work by Gu, Kelly, and Xiu (2020) demonstrates that ML models, particularly neural networks, significantly outperform linear methods by efficiently modeling non-linear interactions among a high-dimensional set

of firm characteristics. This capability suggests that ML is not merely a superior predictive tool for returns but also a powerful instrument for processing fundamental data itself. Consequently, a compelling opportunity exists to employ ML to generate forecasted fundamental measures. By using a model like an MLP to predict future fundamentals directly from their past values, we can create more timely and potentially more accurate input signals. This approach moves beyond using raw, historical data and aims to provide a purified, forward-looking information set for subsequent asset pricing models.

The efficacy of any predictive model in asset pricing is contingent upon the economic relevance of its input features. The selection of fundamental characteristics in this study is grounded in the comprehensive empirical findings of Freyberger, Neuhierl and Weber (2020) and Green, Hand, and Zhang (2017), who systematically identified a parsimonious set of firm attributes that deliver independent information about average U.S. monthly stock returns. These characteristics, spanning dimensions such as value, profitability, investment, and accruals, have been rigorously validated as persistent drivers of cross-sectional returns. By building upon this established set of features, our research ensures that the model inputs are economically meaningful and empirically robust. This allows us to focus the investigation not on feature selection, but on the core methodological question: whether forecasting these fundamentals non-linearly in advance provides a significant performance advantage over simply using their lagged values in a direct prediction model.

3 Research Methodology

3.1 Data Source and Processing

The research utilizes a comprehensive dataset spanning from January 2014 to December 2023. The primary data components are as follows:

1. Profitability Factors:

- **ROE** (`f_profit`): operating profit to book value TTM.
- **ATN**: Accrual to net profit TTM

2. Valuation Factors:

- **BOOK**: Book value of equity per share MRQ.
- **EPS**: Earnings per share MRA.
- **F_VALUE**: Book-to-Market Ratio.

3. Investment and Growth Factors:

- **AST_GRTH**: Growth rate of total assets MRQ.
- **CAP_GAIN**: Capital gains over past 252 trading days.
- **OP_AST_CNG**: Change in net operating assets.

4. Market and Size Factors:

- **F_MKT**: Equal-weighted average return of all stocks at time t.
- **F_SIZE**: Market capitalization (`mcap`) after cross-sectional standardization.

5. Momentum and Liquidity Factors:

- **MOM**: Cumulative return over the past 21 trading days.
- **AMT**: Average daily trading volume over the past 21 trading days.
- **TURN**: Average daily share turnover over the past 252 trading days.

Raw financial data often contains missing entries and extreme outliers, which can severely distort statistical analysis and machine learning model training. Therefore, for each factor and at each time period t, this research replace any missing value for a stock with the **cross-sectional median** of all available values for that factor at time t.

To ensure that factors are comparable across time and that their scales do not unduly influence the neural network models, each factor is cross-sectionally standardized at every time period.

Prior to model input, a final cleaning step removes any remaining infinite or negative infinite values that may have arisen during calculations (e.g., division by zero in factor construction like Book-to-Market), ensuring the integrity of the input matrices for both the linear OLS and nonlinear MLP estimators.

3.2 Base Hypothesis Testing

This study first establishes a foundational hypothesis to validate the timeliness and relevance of fundamental information in asset pricing:

H1: *Contemporaneous fundamental factors exhibit significantly stronger linear correlation with contemporaneous excess returns than lagged fundamental factors.*

To test H1, we construct a comparative linear regression framework based on the Fama-French five-factor model:

Model A (Lagged Model):

$$R_{t+1} = \alpha + \beta_1 \text{MKT}_t + \beta_2 \text{SIZE}_t + \beta_3 \text{VALUE}_t + \beta_4 \text{PROFIT}_t + \beta_5 \text{INVEST}_t + \varepsilon_t \tag{1}$$

Model B (Contemporaneous Model):

$$R_{t+1} = \alpha + \beta_1 \text{MKT}_t + \beta_2 \text{SIZE}_t + \beta_3 \text{VALUE}_{t+1} + \beta_4 \text{PROFIT}_{t+1} + \beta_5 \text{INVEST}_{t+1} + \varepsilon_{t+1} \tag{2}$$

Where:

- R_t: Stock excess returns at period t
- MKT_t: Market factor at period t (constructed as cross-sectional average returns)
- SIZE, VALUE, PROFIT, INVEST: Size, value, profitability, and investment factors

The key difference of these two models is the time period of fundamental factors (VALUE, PROFIT, INVEST). We test the H1 by comparing the in-sample R-squared, post-sample IC mean, post-sample ICIR and the statistical significance of size, value, profitability, and investment factors.

3.3 Fundamental Prediction and Model Structure

3.3.1 Baseline Model

The baseline model serves as a performance benchmark and follows a conventional supervised learning paradigm. Its objective is to establish a direct mapping from a comprehensive set of observable factors at time t to the cross-section of stock excess returns at time $t+1$. The design philosophy prioritizes simplicity and establishes a lower bound for predictive performance against which more sophisticated architectures can be evaluated.

The baseline model is implemented as a Multilayer Perceptron (MLP) and can be formally expressed as:

$$\hat{R}_{t+1} = \text{MLP}_{\text{outer}}\left(\mathbf{F}_t, \mathbf{C}_t; \mathbf{\Theta}_{\text{base}}\right) \tag{3}$$

where:

- $\hat{R}_{t+1}$ is the vector of predicted excess returns for all stocks at period $t+1$,
- $\mathbf{F}_t = [\text{ROE}_t, \text{ATN}_t, \text{BOOK}_t, \text{EPS}_t, \text{AST_GRTH}_t, \text{CAP_GAIN}_t, \text{OP_AST_CNG}_t]$ represents the set of seven fundamental factors,
- $\mathbf{C}_t = [\text{MKT}_t, \text{MCAP}_t, \text{MOM}_t, \text{AMT}_t, \text{TURN}_t]$ represents the set of five non-fundamental control factors (market, size, momentum, amount, and turnover),
- $\mathbf{\Theta}_{\text{base}}$ denotes the model's parameters, including layer size, activation function and regularization.

3.3.2 Two-Stage Model

The two-stage model addresses the inherent information lag in financial reporting by decoupling the prediction task. Its core innovation is an intermediate stage that forecasts future fundamental factors, thereby generating more timely and potentially less noisy input features for the final return prediction. The philosophy is that predicting fundamentals is a more structured and learnable task than directly predicting noisy returns, and that enhanced inputs should lead to enhanced outputs.

The first stage is dedicated to predicting the next period's fundamental factors using only their current values, implemented as a multi-output regression MLP:

$$\hat{\mathbf{F}}_{t+1} = \mathrm{MLP}_{\mathrm{inner}}\left(\mathbf{F}_t; \mathbf{\Theta}_{\mathrm{inner}}\right) \tag{4}$$

where $\hat{\mathbf{F}}_{t+1}$ is the predicted vector of all seven fundamental factors for period $t+1$. This stage is trained using Mean Squared Error (MSE) loss, as the goal is accurate numerical prediction of the factor values themselves.

The second stage mirrors the architecture of the baseline model but uses the *predicted* fundamentals from Stage I as inputs:

$$\hat{R}_{t+1} = \mathrm{MLP}_{\mathrm{outer}}\left(\hat{\mathbf{F}}_{t+1}, \mathbf{C}_t; \mathbf{\Theta}_{\mathrm{outer}}\right) \tag{5}$$

Crucially, the architecture of $\mathrm{MLP}_{\mathrm{outer}}$ is identical to the baseline model's $\mathrm{MLP}_{\mathrm{outer}}$. This ensures that any performance difference can be attributed solely to the quality of the input features ($\hat{\mathbf{F}}_{t+1}$ vs. $\mathbf{F}_t$) rather than to differences in the final predictor's capacity or training.

3.4 Training and Evaluation

The entire sample period (January 2014 to December 2023) is partitioned into three distinct with rate 2:6:2, in order to ensure robust model evaluation and prevent data snooping bias.

- **Pre-sample Period (Jan 2014 – Dec 2015)**: This initial period is used for preliminary model validation, architecture selection, and hyperparameter tuning for the MLP models. Performance metrics here provide an early indication of model viability without contaminating the main training set.
- **In-sample Period (Jan 2016 – Dec 2021)**: This constitutes the primary training set. All models are trained exclusively on data from this period. This interval is long enough to capture market variations and provide sufficient data for training complex nonlinear models.
- **Post-sample Period (Jan 2022 – Dec 2023)**: This period is reserved for out-of-sample (OOS) testing. No model parameters are estimated or tuned using data from this interval. It serves as the ultimate test of a model's predictive power.

Both stages of the models utilize fully connected Multilayer Perceptrons (MLPs). Key hyperparameters are as follows:

- **Optimizer**: Adam optimizer with a fixed learning rate of 0.001.
- **Batch Size**: 32 for stochastic gradient descent.
- **Activation Function**: Rectified Linear Unit (ReLU) in all hidden layers.
- **Regularization**: L2 weight regularization with an alpha parameter of 0.0001 is applied to prevent overfitting.
- **Early Stopping**: Training is monitored on a 10% validation split created from the in-sample data; training halts if validation loss does not improve for 10 consecutive epochs.

Specific Model Configurations:

- **Baseline/Single-Stage Model**: A three-hidden-layer MLP with structure (20, 25, 20), trained for 5 epochs.
- **Two-Stage Model**:

1. **Stage I (Inner MLP)**: To predict next-period fundamentals with structure (10, 20, 10) network, trained for 4 epochs with Mean Squared Error (MSE) loss.
2. **Stage II (Outer MLP)**: Uses an identical architecture to the Baseline model (20, 25, 20), trained for 5 epochs. This ensures a fair comparison, isolating performance differences to the input feature quality.

The core evaluation is based on the time series of monthly cross-sectional ICs.

- **IC Mean**: The average monthly IC measures the model's *directional accuracy*—its ability to rank stocks correctly.
- **IC Standard Deviation**: The standard deviation of monthly ICs measures the model's *stability* and consistency of predictive signal.
- **Information Coefficient Information Ratio (ICIR)**: Calculated as IC Mean/IC Std. This is the key risk-adjusted performance metric. A higher ICIR indicates a more reliable and efficient predictive signal.

These statistics are calculated separately for the Pre-sample, In-sample, and critically, the Post-sample periods to assess training fit, tuning efficacy, and OOS robustness, respectively.

4 Empirical Results

4.1 Feasibility of Fundamental Prediction: Non-lagged Fundamentals Perform Better Explain Power

Table 1 presents the ordinary least squares (OLS) regression results for both Model A (Lagged Model) and Model B (Contemporaneous Model). The results demonstrate the superior explanatory power of contemporaneous fundamental factors compared to their lagged counterparts.

Table 1: OLS Regression Results: Lagged vs. Contemporaneous Fundamental Factors

Variables	**Model A (Lagged) Coefficient (t)**	**Model B (Contemporaneous) Coefficient (t)**
Constant	0.0027***	0.0189***
	(6.445)	(45.688)
MKT	0.0953***	0.0892***
	(22.198)	(20.935)
SIZE	-0.0004	0.0008**
	(-1.591)	(3.101)
VALUE	0.0102***	-0.0272***
	(14.026)	(-37.724)
PROFIT	0.0000	0.0000
	(1.070)	(1.308)
INVEST	-0.0001**	-0.0001**
	(-2.093)	(-2.322)
Observations	242,679	243,738
R-squared	0.003	0.008
Prob (F-statistic)	1.42e-146	0.000

* p <0.10, ** p <0.05, *** p <0.01

Standard errors in parentheses

The regression results reveal several important findings supporting the superiority of contemporaneous fundamental data:

Model B has a higher R-squared than model A, indicating that contemporaneous fundamentals explain a larger proportion of the variance in excess returns. The F-statistic of Model B (379.1) is also markedly higher than that of Model A (137.9), confirming the superior statistical significance of the contemporaneous specification.

Particularly, the sign reversal and increased significance of VALUE factor suggest that contemporaneous value has stronger predictive power in terms of stock returns.

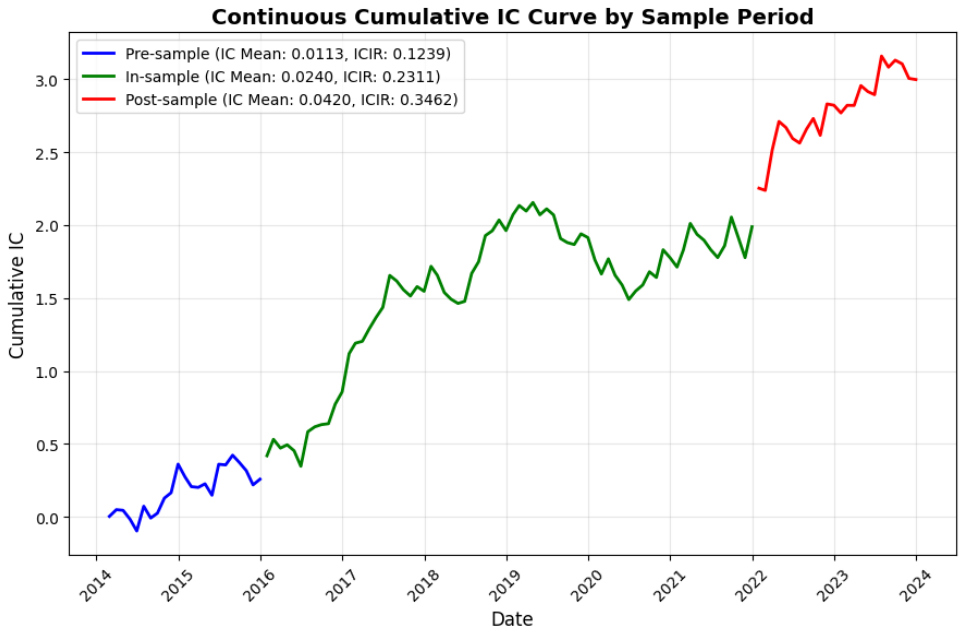

Figure 1: Continuous Cumulative IC Curve for Model A (Lagged Model)

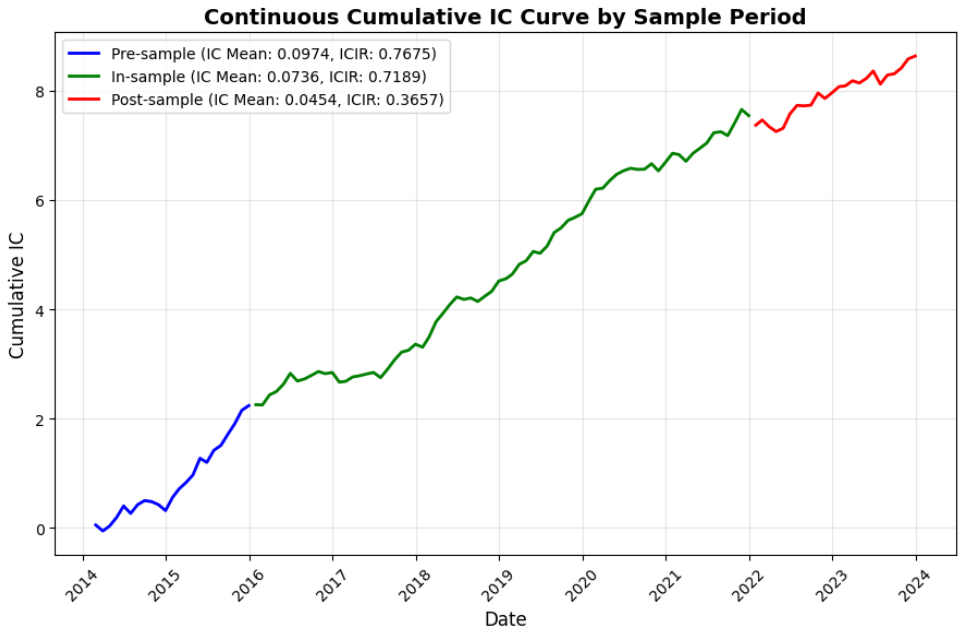

Figure 2: Continuous Cumulative IC Curve for Model B (Contemporaneous Model)

The in-sample Information Coefficient IR (ICIR) provides further evidence of Model B's superiority. The ICIR for Model B (0.7189) is approximately three times higher than that of Model A (0.2311), indicating that contemporaneous factors provide substantially better discriminatory power in forecasting returns.

4.2 Performance Comparison: Two-stage Model Out Perform the Baseline

As shown in IC figures, the Two-Stage model has superior post-sample performance. Its Post-sample IC Mean (0.0611) is **39.2% higher** than the Baseline's (0.0439), also, its Post-sample ICIR (0.5963) is

higher. This indicates that the Two-Stage model provides not only more accurate directional forecasts but also more consistent performance out of sample.

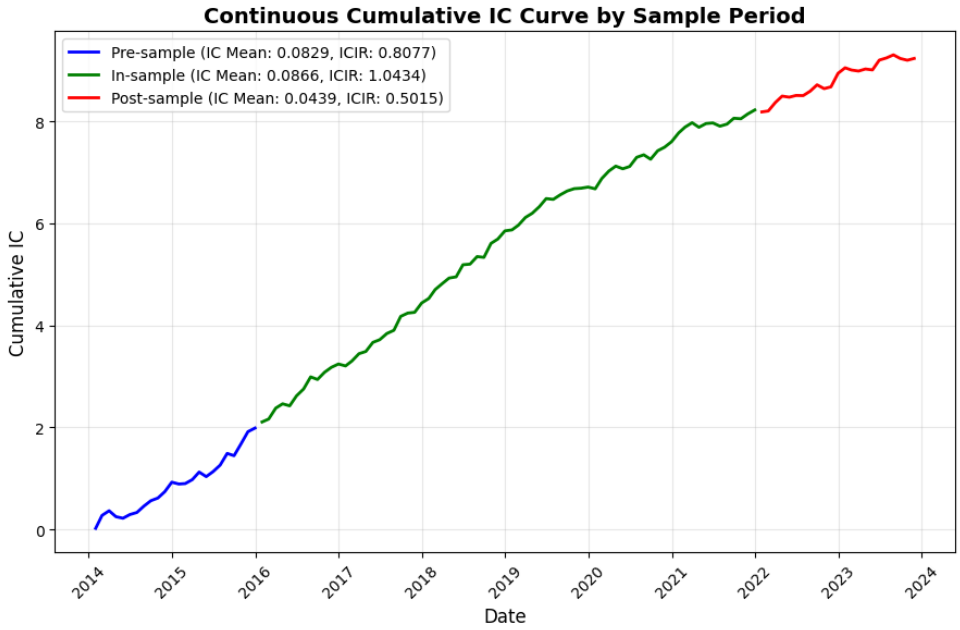

Figure 3: Continuous Cumulative IC Curve for Baseline Model

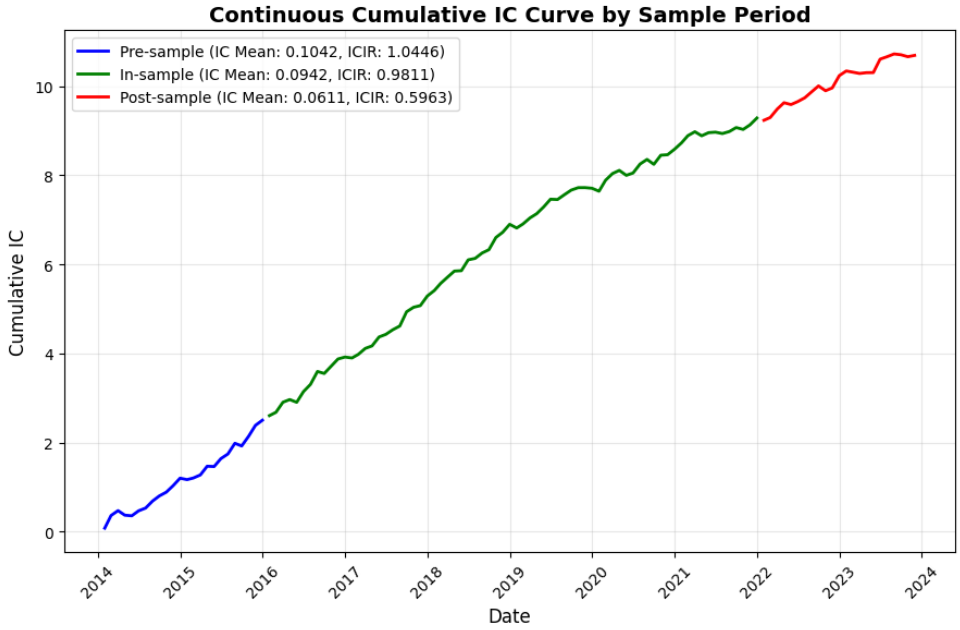

Figure 4: Continuous Cumulative IC Curve for Two-stage Model

This study conducted grouped backtesting on the predicted returns of these two models respectively, plotting the grouped return curves where labels 1 to 5 correspond to predicted returns from low to high. The visualized output results are shown in Figures 5 and 6.

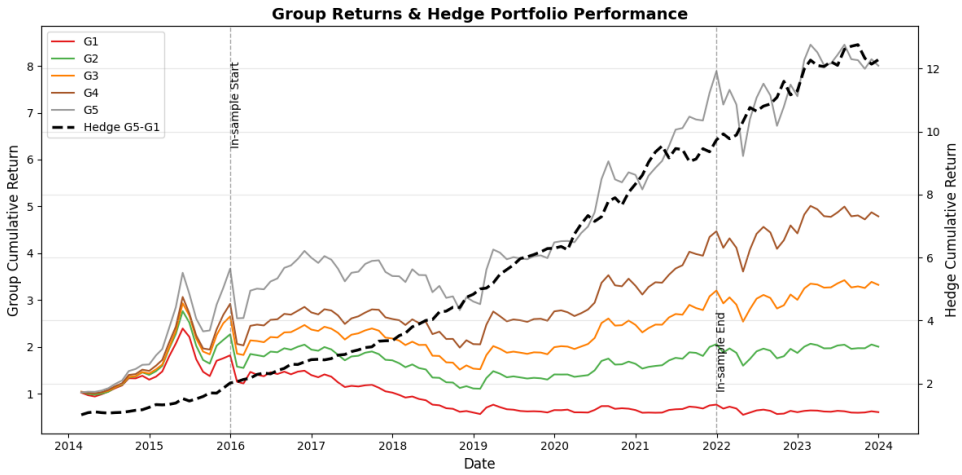

Figure 5: Group Returns & Hedge Portfolio Performance for Baseline Model

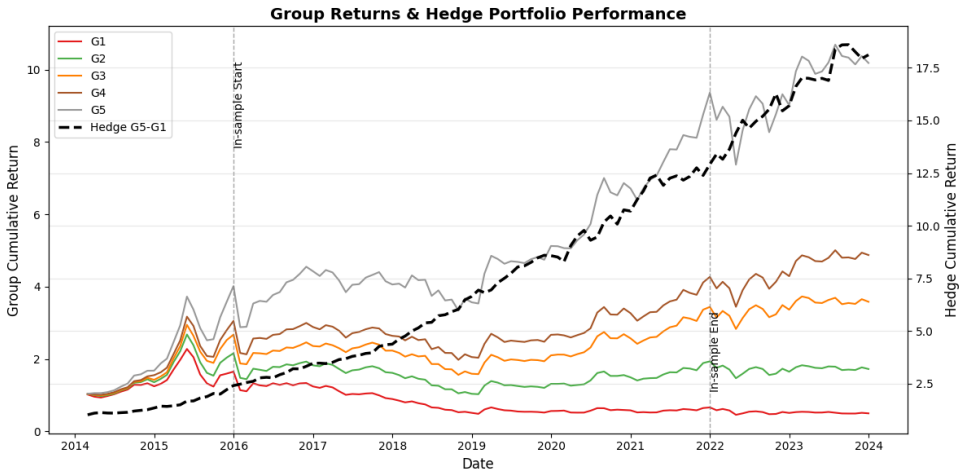

Figure 6: Group Returns & Hedge Portfolio Performance for Two-stage Model

As shown in the figures, the Two-Stage model's hedge portfolio achieves a higher terminal cumulative return. In both figures, the portfolios G1 through G5 are clearly separated in ascending order. The Two-Stage model typically demonstrates wider and more consistent spreads between adjacent quintiles, especially in the post-sample period. This indicates its predicted rankings possess stronger discriminative power across the entire cross-section of stocks.

Table 2 shows that the Two-Stage model generates a higher Sharpe ratio in Post-sample period, reflecting its ability to generate higher realized risk-adjusted returns in a simulated live trading environment.

Therefore, the empirical results conclusively demonstrate that using a nonlinear MLP to forecast fundamental factors can enhance asset pricing model performance, as the intermediate step of fundamental prediction provides a significant information gain, leading to more accurate prediction of return.

Table 2: Hedge Portfolio (G5-G1) Sharpe Ratio Comparison

Model	**Pre-sample**	**In-sample**	**Post-sample**	**Total**
Baseline	2.752	3.154	1.381	2.606
Two-Stage	3.527	2.950	**1.625**	**2.733**

4.3 Mechanism Test: Isolating and Interpreting the Model's Incremental Alpha

The core innovation of the two-stage model lies in its attempt to mitigate the information lag inherent in traditional accounting data by predicting future firm fundamentals (e.g.,$\hat{\mathbf{F}}_{t+1}$). Previous empirical results demonstrate the effectiveness of this core innovation. However, the enhanced performance of our two-stage model over the baseline could arise from two distinct sources: it may simply more efficient capture the interaction of lagged data, or it may generate a genuine, uncorrelated alpha by predicting future fundamental.

5 Conclusion

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